formal charge of $O_A / O_B = N_V - \left(N_e + \frac{N_b}{2}\right)$

$= 6 - \left(4 + \frac{4}{2}\right)$

$= 6 - 6 = 0$

formal charge of C $= 4 - \left(0 + \frac{8}{2}\right)$

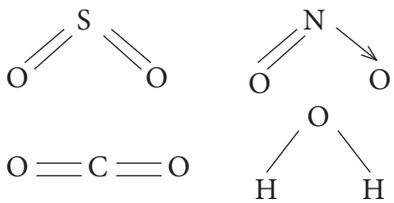
$= 4 - 4 = 0$

$\therefore$ Option (d) – (0, 0, 0)

3. $\ddot{N}H_3$, $\ddot{P}H_3$ - electron rich,
$CH_3 - CH_3$ – Covalent neutral molecule.
BH_3 - electron deficient

option (c)

4.



water contain only σ bonds and no π bonds

option (d) is correct

5. $CH_3 - C \equiv C - C(=O) - H$

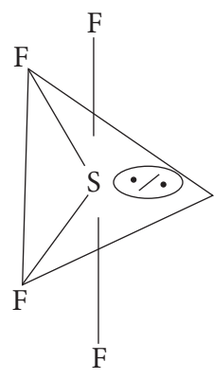
2 – butynal

no. of σ bonds = 8[4C-H;,3C-C;1C-O]

no. of π bonds = 3[2C-C; 1C-O]

$\therefore$ ratio = $\frac{8}{3}$

6.



Normal bond angle in regular trigonal bipyramidal are 90° and 120°

due to *l*.p - b.p repulsion, bond angle reduced 89°, 117°

option (d)

Correct Statement :

7. Oxygen molecule is paramagnetic

Correct Reason

It has two unpaired electron in its antibonding molecular orbital.

Option (c) assertion true and reason false.

8. Option (b) - two half filled orbitals overlap.

9. ClF_3 – Sp^3d hybridisation

NF_3 - Sp^3 hybridisation

BF_3 - Sp^2 hybridisation

Option (d) is correct.

10. Option (b) is correct Sp^3 hybridisation.

Oribital geometry tetrahedron, bond angle 109°28'

11. bond order = $\frac{1}{2}(n_b - n_a)$

bond order of $O_2^{2-} = \frac{1}{2}(8 - 6) = 1$

bond order of $C_2^{+} = \frac{1}{2}(5 - 2) = 1.5$

bond order of $O_2 = \frac{1}{2}(8 - 4) = 2$

bond order of $C_2^{2-} = \frac{1}{2}(8 - 2) = 3$

12. PCl_5 – Sp^3d hybridisation

S, P_x, P_y, P_z and $d_{x^2-y^2}$

option (c) is correct.

13. Option (b) $O_2 > O_3 > H_2O_2$

$2 > 1.5 > 1$

Key answers -jagan.indd 304 22-01-2021 13:23:35